

2920/301
DATA COMMUNICATION
AND NETWORKING
November 2021
Time: 3 Hours



THE KENYA NATIONAL EXAMINATIONS COUNCIL
DIPLOMA IN INFORMATION COMMUNICATION TECHNOLOGY
MODULE III
DATA COMMUNICATION AND NETWORKING
3 hours

INSTRUCTIONS TO CANDIDATES

*This paper consists of EIGHT questions.
Answer any FIVE of the EIGHT questions in the answer booklet provided.
Candidates should answer the questions in English.*

This paper consists of 4 printed pages.

Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.

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Turn over

1.
 - (a) Differentiate between *analog* and *digital* signals as used in data communication. (4 marks)
 - (b) Explain **three** functions of physical layer in OSI model. (6 marks)
 - (c) User Datagram Protocol header is made up of several fields. Explain **three** fields that could be found in the header. (6 marks)
 - (d) Mary intends to use command line network operating system for her computer network. Outline **four** advantages of the operating system that could have influenced her choice. (4 marks)

 2.
 - (a) Outline **four** types of communication software that could be found in a computer. (4 marks)
video conferencing
video chat
 - (b) For each of the following scenarios, suggest the most appropriate Internet connection justifying your answer:
 - (i) a host server that is accessed by millions of users; (1 mark)
 - (ii) a home office that uses existing telephone lines; (1 mark)
 - (iii) a video chat on a mobile phone; (1 mark)
 - (iv) connecting several branches of company located in various towns. (1 mark)
 - (c)
 - (i) Outline **two** uses of a network baseline during network troubleshooting. (2 marks)
 - (ii) Unipolar encoding scheme has several problems that makes it unpopular in modern digital transmission schemes. Explain **two** such problems. (4 marks)
 - (d)
 - (i) Outline **two** roles of Simple Mail Transfer Protocol (SMTP). (2 marks)
 - (ii) Figure 1 shows a copper twisted pair cable (STP). Use it to answer the question that follows:
- The diagram shows a cross-section of a shielded twisted pair (STP) cable. It is divided into four labeled parts: (i) points to the outermost shield, (ii) points to the braided shield layer, (iii) points to one of the four twisted pairs of conductors, and (iv) points to a single conductor within one of the twisted pairs.
- Figure 1
- Outline the functions of the part labelled (i), (ii), (iii) and (iv). (4 marks)
3.
 - (a) Describe **three** organizations that regulates WLAN. (6 marks)
 - (b) Kate designed a computer network that used hybrid network topology. Outline **four** factors that could have influenced her design. (4 marks)
 - (c) Differentiate between *next- generation firewall* and *cloud firewall*. (4 marks)
 - (d) Dynamic Host Configuration Protocol (DHCP) is used to automate IP addressing in a network. Explain **three** components of this protocol. (6 marks)

4. (a) Explain each of the following terms as used in the Internet:
- (i) attachment; (2 marks)
 - (ii) favourite. (2 marks)
- (b) Peter was advised by his friend to use unattended installation for a network software. Outline **four** advantages of this installation method. (4 marks)
- (c) Judy discovered several network vulnerabilities in her company's network. Outline **four** examples of these vulnerabilities that she could have noted. (4 marks)
- (d) (i) Alice intends to use WiMAX (Worldwide Interoperability for Microwave Access) to interconnect the two network branches located in a town. Explain **two** features of this transmission media that could have influenced her decision. (4 marks)
- (ii) Explain **two** challenges of a bus network topology. (4 marks)
5. (a) (i) Outline **three** challenges of a peer to peer network model. (3 marks)
- (ii) Distinguish between a *hub* and a *switch*. (4 marks)
- (b) (i) Outline **three** factors that determines the bandwidth of a transmission media. (3 marks)
- (ii) Betty intends to troubleshoot a computer network for a client. Outline **four** common hardware troubleshooting tools that she could use. (4 marks)
- (c) During network design, networks are categorized based on the number of nodes connected. Explain **three** such categories. (6 marks)
6. (a) Outline two parts of Ethernet MAC address. (2 marks)
- (b) For each of the following functions of Ethernet frame, identify the field responsible:
- (i) detects errors in an Ethernet frame; (1 mark)
 - (ii) describes which higher-level protocol is used; (1 mark)
 - (iii) notifies the destination to get ready; (1 mark)
 - (iv) shows frame MAC address of the origin. (1 mark)
- (c) Carrier Sense Media Access/Collision Detection uses several steps that guarantee delivery of data packet from sender to receiver host. Outline **four** such steps. (4 marks)
- (d) (i) Outline **four** factors that should be considered before installing communication software in a computer. (4 marks)
- (ii) Margret was diagnosing a tablet that was not connecting to the company's WIFI. Outline **three** possible causes and solutions for the problem. (6 marks)
7. (a) Describe **two** types of fibre-optic connectors. (4 marks)
- (b) (i) Outline **two** disadvantages of time-division multiplexing scheme (TDM). (2 marks)
- (ii) Distinguish between *access rate* and *committed information rate* as used in frame relay configurations. (4 marks)

- (c) Joan was required to install a computer network to use the most current technology in data communication. Explain **two** challenges that she could have experienced during the installation. (4 marks)
- (d) Internet Service Providers (ISP) prefer using Point-to-Point protocol when connecting their customers to WAN. Explain **three** advantages that could have influenced the trend. (6 marks)

8. (a) (i) Outline **two** characteristics of WAN. (2 marks)
- (ii) Distinguish between *unicast* IPv6 address and *anycast* IPv6 address. (4 marks)
- (b) Paul noted that Cyclic Redundancy Check (CRC) is an error-detecting scheme that is commonly used in data communication systems. Outline **four** advantages that could be influencing this trend. (4 marks)
- (c) Every Ethernet uses Address Resolution Protocol for it to function properly. Explain **two** functions of this protocol. (4 marks)
- (d) Figure 2 shows a TCP/IP model. Use it to answer the questions that follows.

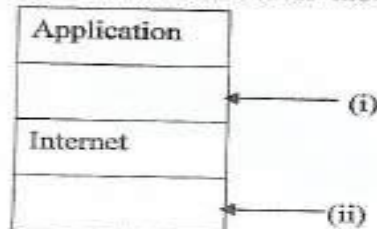


Figure 2

- (i) Explain the function of the layers labelled (i) and (ii). (4 marks)
- (ii) List **four** protocols that could be used in the application layer. (2 marks)

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